

1. BASIC SYMPLECTIC GEOMETRY

There are essentially only two standard textbooks in symplectic geometry by McDuff-Salamon: *Introduction to Symplectic Geometry* and *J-holomorphic Curves and Symplectic Topology*. But there are some alternatives now, as pointed out by Prof. Zhengyi Zhou.

If one finds the learning curve of McDuff-Salamon to be too steep, it is usually recommended that one first goes through Ana Cannas da Silva's **Lectures on Symplectic geometry**. This book goes through moment maps and symplectic toric geometry pretty carefully, which is also important for the study of symplectomorphism groups and is not found in McDuff-Salamon. There are also many newer online lecture notes, such as [Michael Usher's notes](#) (basic symplectic geometric notions explained carefully), [Denis Auroux's notes](#) (Hand-written notes towards Lagrangian Floer theories) or [Jonny Evans's notes](#) (which is my favorite introductory notes, and it points towards applications on symplectomorphism groups).

On top of that, Prof. Zhengyi Zhou has recommended in his document on Geiges-Zehmisch's **A course on holomorphic discs** and other great references such as Wendl's lecture notes as a better introduction to holomorphic curve theory.

After this point, there are several existing approaches to the topology of symplectomorphism groups of 4-manifolds: **holomorphic foliations, inflation techniques, and ball-swapping techniques**.

2. FOLIATION METHODS

The first reference to holomorphic foliations is Gromov's original paper [Pseudo holomorphic curves in symplectic manifolds](#). Hofer-Wysocki-Zehnder's paper [Finite energy foliations of tight three-spheres and Hamiltonian dynamics](#) is not directly tied to symplectomorphism groups but is still a worthy reading on this topic.

Neither reference is very easy to read for beginners. Eliashberg and Polterovich's [Local Lagrangian 2-Knots are Trivial](#) is a great introduction of using holomorphic foliations towards Lagrangian uniqueness. Richard Hind's [Lagrangian spheres in \$S^2 \times S^2\$](#) serves as a great showcase of how foliation techniques are used in combination of symplectic field theory. Rizell-Goodman-Ivrii's recent paper [Lagrangian isotopy of tori in \$S^2 \times S^2\$ and \$\mathbb{C}P^2\$](#) represents the state-of-art in this direction.

3. INFLATION TECHNIQUES AND TOPOLOGY OF ALMOST COMPLEX STRUCTURE

In my opinion, the best introductory material for the inflation technique is McDuff's [From deformation to isotopy](#). This paper includes a very careful proof of the connectedness of symplectic ball-packings in symplectic manifolds with $b^+ = 1$ using the simplest version of inflation techniques. It also contains a non-vanishing GW result using Taubes' famous GW=SW theorem, showcasing the combination of the two directions.

The inflation techniques must be coupled with understandings of almost complex structures for the study of symplectomorphism groups. The relation between the topology of symplectomorphism groups and almost complex structures which allows negative curves were studied in the series of works of Abreu's [Topology of symplectomorphism groups of \$S^2 \times S^2\$](#) and

Abreu-McDuff's [Topology of symplectomorphism groups of rational ruled surfaces](#), which should be the start of learning techniques in this direction.

Li-Usher's construction of negative inflation [Symplectic forms and surfaces of negative square](#) plays an important role in more complicated situations of the study of symplectomorphism groups. McDuff's [Symplectomorphism groups and almost complex structures](#) shows the landscape of how inflations can be paired with the study of almost complex structures. Olguta Buse's paper [Negative inflation and stability in symplectomorphism groups of ruled surfaces](#) contains showcase results of how one may use negative inflations to study the space of almost complex structures and symplectomorphism groups.

The state-of-art of symplectic/almost complex inflations is McDuff-Opstein's [Nongeneric J-holomorphic curves and singular inflation](#). This paper covers most applications which involves configurations of negative curves.

4. BALL-SWAPPING TECHNIQUES

The idea of symplectic ball-swapping comes from algebraic geometry and is first explained in Seidel's excellent paper [Lectures on four dimensional Dehn twists](#). This paper requires some understandings on the configuration space of points on an algebraic surface, but is essentially self-contained (but it takes some time for readers unfamiliar to algebraic geometry to get used to the language therein). It is important to understand that ball-swapping is a symplectic incarnation of a type of monodromy map in the moduli space of the corresponding Kähler variety.

The implementation of this idea using ball-packing is first done in [Lagrangian spheres, symplectic surfaces and symplectic mapping class groups](#) by Li-Wu (see a sketch of the proof of Conjecture 1.7 in the final section), which might be easier for those with no background in algebraic geometry. The key point is to consider it as a symplectomorphism which comes from a loop of ball-embeddings.

The conjecture is proved in Borman-Li-Wu [Spherical Lagrangians via ball packings and symplectic cutting](#) carrying out this idea. The *ball-swapping symplectomorphism* is formalized in [Exact Lagrangians in \$A_n\$ -surface singularities](#), and defined also in the non-compact setting.

Ball-swappings are applied to studied the connectedness of symplectomorphism groups and Lagrangian Dehn twists in [Symplectic 2 spheres and the symplectomorphism group of small rational 4-manifolds](#) and [Symplectic Torelli groups of rational surfaces](#).

Other useful references include Evans' papers [Symplectic mapping class groups of some Stein and rational surfaces](#), [Lagrangian spheres in Del Pezzo surfaces](#); and a recent trend of reviewing the symplectomorphism groups using Seiberg-Witten theory [From flops to diffeomorphism groups](#).